$\mathcal{ALCI}_{\mathsf{RCC5}}$ under Strong EQ: Corrected Local Normalization and Explicit Depth-Indexed Signatures

Technical note prepared for discussion

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Abstract

This note continues the earlier blocking draft for $\mathcal{ALCI}_{\mathsf{RCC5}}$ under strong EQ. The previous revision already corrected one false local lemma by replacing flat predecessor classes with coherent predecessor blocks. The present addendum makes the next step explicit: it defines a concrete depth-indexed signature construction and proves the associated finite-index lemma. The construction uses bounded rounds of predecessor-color refinement over the older neighborhood of a tableau node, together with multiplicity tables, RCC5 relation-set tables, and projected witness footprints. The resulting state space is finite at every modal depth. This closes the *finite-index* part of the blocking program. The note remains deliberately careful about scope: the last replay theorem needed for a full decidability claim still has to be written against this explicit signature system.

Status after this addendum

Two things are now explicit.

- The local combinatorial obstruction has been repaired: coherent-block normalization is proved.
- The depth-indexed cache alphabet is no longer schematic: it is now given by an explicit bounded refinement construction, and its finite-index lemma is proved.

What is *not* claimed here is the final logic-level decidability theorem. The remaining step is a replay theorem showing that the projected witness footprints carried by the explicit signatures suffice for blocked unraveling.

1 Introduction

Wessel's report isolates the missing step for $\mathcal{ALCI}_{\mathsf{RCC5}}$: the prototype tableau may be taken as sound and complete at the local rule level, but there is no blocking or infinity-checking argument that turns it into a terminating decision procedure. The reason is structural. Models are complete graphs rather than trees, so every older node constrains every fresh witness immediately, and PP/PPI may force genuine infinitary unfolding.

The previous draft note addressed the local part of this issue, but one classwise witness-normalization lemma was too strong. The revised version corrected that by proving normalization on *coherent*

predecessor blocks. In the previous exchange this left one next step: define the finite depth-indexed signature system explicitly and prove that only finitely many such signatures can occur for a fixed input concept.

This note does exactly that. Its contribution is therefore narrower than a full decidability proof, but sharper than an informal design sketch.

2 Preliminaries

We work in $\mathcal{ALCI}_{\mathsf{RCC5}}$ under the strong-EQ semantics. Distinct nodes are connected by exactly one of the four non-EQ RCC5 base relations

$$\mathcal{R} = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\},$$

with DR and PO symmetric, PP and PPI converse to one another, and $\mathsf{PP}, \mathsf{PPI}$ strict partial orders.

Fix an input concept C in negation normal form, with closure $\mathrm{cl}(()C)$ and modal depth $d = \mathrm{md}(C)$. For each $k \leq d$, write

$$\mathrm{cl}_k(C) = \{D \in \mathrm{cl}(()C) \mid \mathrm{md}(D) \leq k\}.$$

For a saturated tableau node x, define the truncated local type

$$\mathrm{tp}_k(x) = \lambda(x) \cap \mathrm{cl}_k(C).$$

If $R \in \mathcal{R}$, the R-payload of a type t is

$$\Gamma_R(t) = \{D \in \mathrm{cl}(()C) \mid \forall R.D \in t\}.$$

We assume an underlying unblocked tableau branch consisting of:

- a finite set of current branch nodes,
- a creation order $\prec$,
- a saturated type $\lambda(x)$ for each node x,
- a unique base relation $\rho(x, y) \in \mathcal{R}$ for each ordered pair of distinct nodes.

For a node x, let

$$\mathsf{old}(x) = \{u \mid u \prec x\}$$

be its older neighborhood.

Definition 2.1 (Triangle predicate)**.** For $R, S, T \in \mathcal{R}$, write

$$\mathrm{Tri}(R, S, T)$$

if and only if the RCC5 composition table allows the oriented triangle

$$R(a, b), \qquad S(b, c), \qquad T(a, c).$$

Equivalently, $T \in R \circ S$.

Thus, if u, v are older nodes and y is a fresh witness with

$$\rho(u, v) = q, \qquad \eta(v) = s, \qquad \eta(u) = r,$$

then local admissibility requires $\mathrm{Tri}(q, s, r)$.

3 Symbolic one-point extensions

Because models are complete graphs, an existential witness is never determined by a single edge. A tableau rule for $\exists S.E$ must describe how the fresh node sits with respect to *all* older nodes.

Definition 3.1 (Locally admissible symbolic extension)**.** Let x be a saturated tableau node and let $\exists S.E \in \lambda(x)$. A *symbolic one-point extension* for this existential is a pair (t, η) where:

(a) t is a saturated target type with $E \in t$ and $\Gamma_S(\lambda(x)) \subseteq t$;

(b) η maps every $u \in \mathsf{old}(x)$ to a role $\eta(u) \in \mathcal{R}$, intended as the label from u to the new witness.

It is *locally admissible* if, in addition,

(i) for every $u \in \mathsf{old}(x)$,

$$\Gamma_{\eta(u)}(\lambda(u)) \subseteq t;$$

(ii) for all distinct $u, v \in \mathsf{old}(x)$,

$$\mathrm{Tri}\big(\rho(u,v), \eta(v), \eta(u)\big)$$

holds.

This symbolic complete-graph witness is the local object that any blocker must replay.

4 Why flat predecessor classes do not suffice

The first correction is negative.

Definition 4.1 (Flat predecessor class)**.** Relative to a fixed node x, two older nodes $u, v \prec x$ are in the same *flat class* if

$$\lambda(u) = \lambda(v), \qquad \rho(u,x) = \rho(v,x), \qquad \Gamma_{\rho(u,x)}(\lambda(u)) = \Gamma_{\rho(v,x)}(\lambda(v)).$$

Proposition 4.2 (Flat classwise normalization is false)**.** *There is a satisfiable local branch configuration with a node x and an admissible symbolic extension (t, η) such that no locally admissible symbolic extension with the same target type is constant on flat predecessor classes.*

Proof. Let x carry the existential demand

$$\exists \mathsf{DR}.(p \sqcap q \sqcap r).$$

Assume the older neighborhood of x consists of nodes

$$a_1, a_2, b_1, b_2, c,$$

all related to x by DR.

Assign local types so that

$$\tau_A : \ \forall \mathsf{PO}.\neg p, \ \forall \mathsf{PP}.\neg p, \ \forall \mathsf{PPI}.\neg p,$$

$$\tau_B : \ \forall \mathsf{PO}.\neg q, \ \forall \mathsf{PP}.\neg q,$$

$$\tau_C : \ \forall \mathsf{DR}.\neg r, \ \forall \mathsf{PP}.\neg r, \ \forall \mathsf{PPI}.\neg r.$$

Take

$$\lambda(a_1) = \lambda(a_2) = \tau_A, \qquad \lambda(b_1) = \lambda(b_2) = \tau_B, \qquad \lambda(c) = \tau_C.$$

Among the older nodes, let every edge be DR except

$$b_2 \ \mathsf{PPI} \ c \qquad \text{equivalently} \qquad c \ \mathsf{PP} \ b_2.$$

Now choose the witness type

$$t = \{p, q, r, \top\}$$

and labels

$$\eta(a_1) = \eta(a_2) = \mathsf{DR}, \qquad \eta(b_1) = \mathsf{DR}, \qquad \eta(b_2) = \mathsf{PPI}, \qquad \eta(c) = \mathsf{PO}.$$

This symbolic extension is locally admissible. The nodes b_1 and b_2 lie in the same flat predecessor class relative to x, but they cannot be uniformized: forcing both to DR breaks the triangle with b_2 PPI c and c PO wit, while forcing both to PPI breaks the triangle with b_1 DR c and c PO wit. Hence flat predecessor classes are too coarse. □

5 Coherent predecessor blocks and the corrected normalization lemma

The right local granularity is obtained by refining the older neighborhood until inter-block edges are uniform.

Definition 5.1 (Coherent partition)**.** Fix a saturated node x. A partition P_x of $\mathsf{old}(x)$ is *coherent* if:

(i) it refines flat predecessor classes;

(ii) for every two distinct blocks $B, C \in P_x$ there is a single role $q_{BC} \in \mathcal{R}$ such that for all $u \in B$ and $v \in C$,

$$\rho(u, v) = q_{BC};$$

(iii) for every block $B \in P_x$ with at least two nodes there is a single internal label $q_B \in \{\mathsf{DR}, \mathsf{PO}\}$ such that for all distinct $u, v \in B$,

$$\rho(u, v) = q_B.$$

Proposition 5.2 (Existence of coherent refinements)**.** *Every finite older neighborhood admits a coherent partition refining the flat predecessor classes.*

Proof. Start from the flat partition. Whenever a block fails coherence, split it according to the offending external or internal edge label. Every split strictly refines a finite partition, so the process terminates. The terminal partition is coherent by construction. □

Lemma 5.3 (Coherent-block normalization)**.** *Let x be a saturated branch node, let P_x be any coherent partition of $\mathsf{old}(x)$, and let (t, η) be a locally admissible symbolic one-point extension for an existential $\exists S.E \in \lambda(x)$. Then there is a locally admissible symbolic one-point extension (t, η') with the same target type such that:*

(a) η' is constant on every block $B \in P_x$;

(b) for every block $B \in P_x$, the common value $\eta'(B)$ is one already realized by some member of B under η.

Proof. For each coherent block B, let

$$L_B = \{\eta(u) \mid u \in B\} \subseteq \mathcal{R}.$$

We choose one value $\eta'(B) \in L_B$ per block.

If B is a singleton, choose its unique realized value.

If $|B| > 1$ and the internal label is PO, then every role is stable on B, because $\mathrm{Tri}(\mathsf{PO}, r, r)$ holds for all $r \in \mathcal{R}$. So any realized value may be chosen.

If $|B| > 1$ and the internal label is DR, then PPI is not stable since $\mathrm{Tri}(\mathsf{DR}, \mathsf{PPI}, \mathsf{PPI})$ fails. Hence the original admissible extension cannot label every member of B by PPI; otherwise two distinct members of B would already violate local admissibility. So some realized label in L_B lies in $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, and each of these is stable on a DR-block. Choose such a stable realized value.

Now let $B \neq C$ be distinct coherent blocks. By coherence there is a single label q_{BC} between them. If we choose $u \in B$ and $v \in C$ with $\eta(u) = \eta'(B)$ and $\eta(v) = \eta'(C)$, then the original local admissibility of (t, η) yields

$$\mathrm{Tri}\big(q_{BC}, \eta'(C), \eta'(B)\big).$$

Thus all inter-block triangles remain valid. Payload conditions are preserved because coherence refines the flat classes, so all nodes in one coherent block have the same local type and hence the same payloads. Therefore (t, η') is again locally admissible. □

6 Explicit depth-indexed signatures

This section makes the depth-indexed construction explicit. The point is to build a *uniform, finite* alphabet of predecessor colors and then package from it a finite signature for each tableau node.

6.1 Bounded predecessor-color refinement

The construction is recursive in the modal depth.

Definition 6.1 (Rank-0 signatures)**.** Let

$$\mathrm{Sig}_0 = \{\mathrm{tp}_0(x) \mid x \text{ is a saturated tableau node}\}.$$

Equivalently, Sig_0 is the set of saturated subsets of the finite propositional closure $\mathrm{cl}_0(C)$. For a node x, put

$$\mathrm{Prof}_0(x) = \mathrm{tp}_0(x).$$

Assume now that $k < d$ and that the finite set Sig_k has already been defined.

Definition 6.2 (Global refinement alphabets)**.** Define finite alphabets $A_{k+1,0}, A_{k+1,1}, \ldots, A_{k+1,k+1}$ recursively by

$$A_{k+1,0} = \mathrm{Sig}_k \times \mathcal{R} \times \mathcal{P}(\mathrm{cl}_k(C)),$$

and, for $0 \le i < k+1$,

$$A_{k+1,i+1} = A_{k+1,i} \times \big(\mathcal{P}(\mathcal{R})\big)^{A_{k+1,i}}.$$

We write

$$A_{k+1} = A_{k+1,k+1}.$$

Remark 6.3. The choice of exactly $k+1$ refinement rounds is deliberate. The earlier notes spoke informally about "stabilizing" a quotient on the older neighborhood. That would depend on the branch and would not by itself give a uniform global alphabet. Here the recursion is bounded by the modal depth, so the alphabet is defined once and for all from the input concept.

Definition 6.4 (Local predecessor colors at rank $k+1$)**.** Fix a saturated node x and an older node $u \in \mathsf{old}(x)$.

The initial predecessor color of u relative to x is

$$\mathrm{Col}^x_{k+1,0}(u) = \big(\mathrm{Prof}_k(u), \rho(u,x), \Gamma_{\rho(u,x)}(\lambda(u)) \cap \mathrm{cl}_k(C)\big) \in A_{k+1,0}.$$

Assume inductively that $\mathrm{Col}^x_{k+1,i}(u) \in A_{k+1,i}$ has been defined for every $u \in \mathsf{old}(x)$. For each such u, define the relation-set profile

$$\Theta^x_{k+1,i}(u) : A_{k+1,i} \to \mathcal{P}(\mathcal{R})$$

by

$$\Theta^x_{k+1,i}(u)(a) = \{\rho(u,v) \mid v \in \mathsf{old}(x),\ v \neq u,\ \mathrm{Col}^x_{k+1,i}(v) = a\}.$$

Then define

$$\mathrm{Col}^x_{k+1,i+1}(u) = \big(\mathrm{Col}^x_{k+1,i}(u), \Theta^x_{k+1,i}(u)\big) \in A_{k+1,i+1}.$$

The final predecessor color of u at rank $k+1$ is

$$\mathrm{Col}^x_{k+1}(u) = \mathrm{Col}^x_{k+1,k+1}(u) \in A_{k+1}.$$

Intuitively, one starts from lower-rank signatures plus incoming edge and payload data, then performs $k+1$ rounds of refinement by looking at which RCC5 relation-sets a predecessor sees towards each current color.

6.2 Context tables and projected witness footprints

The final predecessor colors are now packaged into a finite summary.

Definition 6.5 (Rank-$(k+1)$ context table)**.** Fix a saturated node x.

The rank-$(k+1)$ *multiplicity map* of x is

$$\mu^x_{k+1} : A_{k+1} \to \{0, 1, +\}$$

where $\mu^x_{k+1}(a) = 0$ if no older node has final color a, $\mu^x_{k+1}(a) = 1$ if exactly one older node has that color, and $\mu^x_{k+1}(a) = +$ if at least two older nodes have that color.

The rank-$(k+1)$ *relation-set table* of x is

$$Q^x_{k+1} : A_{k+1} \times A_{k+1} \to \mathcal{P}(\mathcal{R})$$

defined by

$$Q^x_{k+1}(a,b) = \{\rho(u,v) \mid u,v \in \mathsf{old}(x),\ u \neq v,\ \mathrm{Col}^x_{k+1}(u) = a,\ \mathrm{Col}^x_{k+1}(v) = b\}.$$

If there is no such pair (u,v), then $Q^x_{k+1}(a,b) = \varnothing$.

The last finite ingredient is the existential side. Here we do not yet force a functional replay map on final colors. Instead we store the finite *projected footprints* of coherent-block-normalized witnesses.

Definition 6.6 (Projected witness footprint)**.** Fix a saturated node x, let $\exists S.E \in \mathrm{tp}_{k+1}(x)$, and let (t,η) be a coherent-block-normalized symbolic one-point extension for this existential, as in Theorem 5.3. Define its rank-$(k+1)$ projected footprint to be the triple

$$(\exists S.E,\ t \cap \mathrm{cl}_k(C),\ \Lambda),$$

where

$$\Lambda : A_{k+1} \to \mathcal{P}(\mathcal{R})$$

is given by

$$\Lambda(a) = \{\eta(B) \mid B \text{ is a coherent predecessor block of } x \text{ and every } u \in B \text{ satisfies } \mathrm{Col}^x_{k+1}(u) = a\}.$$

Thus $\Lambda(a)$ records which witness labels actually occur on coherent blocks that collapse to the same final predecessor color a.

Definition 6.7 (Explicit rank signatures)**.** Define recursively:

- $\mathrm{Prof}_0(x) = \mathrm{tp}_0(x)$;
- for $k < d$,
$$\mathrm{Prof}_{k+1}(x) = \Big(\mathrm{tp}_{k+1}(x),\ \mu^x_{k+1},\ Q^x_{k+1},\ \mathrm{Req}^x_{k+1}\Big),$$
where Req^x_{k+1} is the set of all projected witness footprints of Theorem 6.6 arising from existential formulas in $\mathrm{tp}_{k+1}(x)$.

Finally let

$$\mathrm{Sig}_{k+1} = \{\mathrm{Prof}_{k+1}(x) \mid x \text{ is a saturated tableau node}\}.$$

Remark 6.8. The use of projected footprints rather than functional witness maps is intentional. It is exactly the point where the note remains conservative. The present section proves that these data are finite. Turning them into the final replay object of the blocker is the remaining step.

7 The finite-index lemma

We now prove that the explicit signature system above has only finitely many states at every rank.

Lemma 7.1 (Finite alphabets at every refinement level)**.** *For every $k < d$ and every $0 \le i \le k+1$, the alphabet $A_{k+1,i}$ from Theorem 6.2 is finite.*

Proof. By induction on i.

For $i = 0$, the set $A_{k+1,0} = \mathrm{Sig}_k \times \mathcal{R} \times \mathcal{P}(\mathrm{cl}_k(C))$ is finite because Sig_k is finite by the outer induction on k, the role set $\mathcal{R}$ is finite, and $\mathrm{cl}_k(C)$ is finite.

Assume $A_{k+1,i}$ finite. Then $\mathcal{P}(\mathcal{R})$ is finite and so is the function space $(\mathcal{P}(\mathcal{R}))^{A_{k+1,i}}$. Hence

$$A_{k+1,i+1} = A_{k+1,i} \times (\mathcal{P}(\mathcal{R}))^{A_{k+1,i}}$$

is finite as well. □

Theorem 7.2 (Explicit finite-index lemma)**.** *For every $k \leq d$, the set Sig_k of explicit rank-k signatures is finite.*

Proof. We argue by induction on k.

Base case $k = 0$. The set Sig_0 is finite because $\mathrm{cl}_0(C)$ is finite and a rank-0 signature is just a saturated subset of that finite set.

Inductive step. Assume Sig_k finite. We show Sig_{k+1} finite.

First, by Theorem 7.1, the global refinement alphabets $A_{k+1,0}, \ldots, A_{k+1}$ are finite.

Second, the truncated type $\mathrm{tp}_{k+1}(x)$ ranges over subsets of the finite set $\mathrm{cl}_{k+1}(C)$, hence only finitely many possibilities exist.

Third, the multiplicity map

$$\mu^x_{k+1} : A_{k+1} \to \{0, 1, +\}$$

ranges over a finite function space because both domain and codomain are finite.

Fourth, the relation-set table

$$Q^x_{k+1} : A_{k+1} \times A_{k+1} \to \mathcal{P}(\mathcal{R})$$

also ranges over a finite function space because A_{k+1} is finite and $\mathcal{P}(\mathcal{R})$ is finite.

Finally, consider the projected witness footprints. A single footprint is determined by:

(a) an existential formula $\exists S.E \in \mathrm{cl}_{k+1}(C)$;

(b) a target k-type $t \cap \mathrm{cl}_k(C)$, of which there are finitely many;

(c) a map $\Lambda : A_{k+1} \to \mathcal{P}(\mathcal{R})$, again chosen from a finite function space.

Hence only finitely many single footprints can exist. Therefore the set Req^x_{k+1} of all footprints occurring at x ranges over the powerset of a finite set, and is itself finite-valued.

Combining the finitely many choices for

$$\mathrm{tp}_{k+1}(x), \qquad \mu^x_{k+1}, \qquad Q^x_{k+1}, \qquad \mathrm{Req}^x_{k+1},$$

we obtain only finitely many possibilities for $\mathrm{Prof}_{k+1}(x)$. Thus Sig_{k+1} is finite. □

Corollary 7.3 (Uniform finite state space)**.** *For a fixed input concept C of modal depth d, the explicit signature construction yields a finite family of state sets*

$$\mathrm{Sig}_0, \mathrm{Sig}_1, \ldots, \mathrm{Sig}_d\,.$$

In particular, any blocking strategy that keys only on these explicit signatures has a uniform finite state space available.

Remark 7.4 (Why this is the right finite-index statement)**.** The theorem above is exactly the step that was previously still schematic. The earlier note spoke about a "finite coherent signature system" in the abstract. Theorem 7.2 now gives one concrete candidate and proves its finiteness directly from the closure of the input and the fixed RCC5 vocabulary.

8 What remains for a full decidability theorem

The present note settles the finite-index part of the blocking program, but it is still deliberately cautious about the final theorem.

The remaining issue is the *replay theorem*. The explicit signatures record projected witness footprints

$$\Lambda : A_{k+1} \to \mathcal{P}(\mathcal{R}),$$

that is, the sets of labels seen on coherent predecessor blocks that collapse to the same final predecessor color. To turn the present note into a full decidability proof one still has to show one of the following:

- either these projected footprints can be sharpened to a functional recipe on final colors without losing realizability;
- or every blocked unraveling step can select coherent representatives from the footprint data in a way that is invariant under equality of signatures.

So the state of play is now more precise than before:

Previous missing step	Current status
Correct local witness normalization	proved via coherent predecessor blocks
Explicit finite depth-indexed cache alphabet	defined and proved finite in Theorem 7.2
Final replay theorem for blocked unraveling	still to be written against the explicit footprint data

This is still substantial progress. The open problem is no longer "find some finite abstraction". One explicit finite abstraction is now on the table. The open task is to prove that its witness component is strong enough for replay.

9 Conclusion

The earlier blocking draft for strong-EQ $\mathcal{ALCI}_{\mathsf{RCC5}}$ needed two ingredients: a correct local normalization principle and a concrete finite signature system. The first was repaired in the previous revision by moving from flat predecessor classes to coherent predecessor blocks. The present note provides the second. It gives an explicit depth-indexed predecessor-color refinement, packages its outcome into multiplicity tables, relation-set tables, and projected witness footprints, and proves that only finitely many such signatures can occur for a fixed input concept.

So one more piece of the decidability program is now explicit and proved: the finite-index lemma. What remains is the replay theorem connecting the projected footprints back to blocked unraveling. Once that step is written out, the route from local tableau soundness/completeness to a terminating decision procedure will be fully specified.

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